CHECK WHAT YOU LEARNT

Check Yourself - Unit 4

A. Choose the correct answer

1. analogRead() gives a number between
   (a) 0 and 1 (b) 0 and 255 (c) 0 and 1023 (d) 0 and 5
2. The LDR module's AO pin must be connected to
   (a) D13 (b) A0 (c) GND only (d) the USB port
3. Serial.begin(9600) is needed so that
   (a) the LED glows (b) the board can talk to the laptop (c) the sensor works (d) code compiles
4. A threshold value should be chosen
   (a) by guessing (b) from real readings (c) from the textbook (d) at random
5. if - else lets a program
   (a) repeat forever (b) make a choice (c) wait (d) measure time
6. A sensor tells the machine
   (a) that it is dark (b) a number (c) what to do (d) the time

B. Fill in the blanks

1 LDR stands for Light ______________ Resistor.

2 The Serial ______________ is the window where you watch live sensor numbers.

3 Tuning a machine to its real surroundings is called ______________.

4 A question in a flowchart is drawn as a ______________.

C. Answer in one or two lines

1 Write the two numbers you measured, bright and dark, and the darkLevel you chose between them.

2 Why is it better to print the sensor value before writing the if - else?

3 Give one real place where an automatic night lamp would save electricity.

4 Your lamp flickers rapidly at dusk. Explain why, and one way to reduce it.

UNIT 5

Final Project: Smart Toll Booth

Sessions 12, 13, 14 and 15

Everything you have learnt now comes together in one machine: a sensor that measures distance, a motor that moves a real barrier, a buzzer, two lamps, and a counter that remembers how many vehicles have passed. This is the project you will demonstrate.

By the end of this unit you will be able to:

- Explain how an ultrasonic sensor measures distance using an echo.
- Control a servo motor to move a physical part to an exact angle.
- Combine one sensor and several outputs in a single program.
- Build, test, present and defend your own working machine.

SESSION 12

one class

Two New Parts and the Project Brief

Today you will

- Learn how the HC-SR04 and the SG90 servo work.
- Understand what your finished toll booth must do.
- Agree the four team roles and start the cardboard build.

The HC-SR04 works exactly like a bat. It sends out a click of sound far too high for human ears, then listens for the echo and times how long it takes to come back. Sound travels about 34 centimetres every millisecond, so the time tells you the distance. The SG90 servo is a different kind of motor: instead of spinning endlessly, you tell it an angle and it turns there and holds position until you say otherwise.

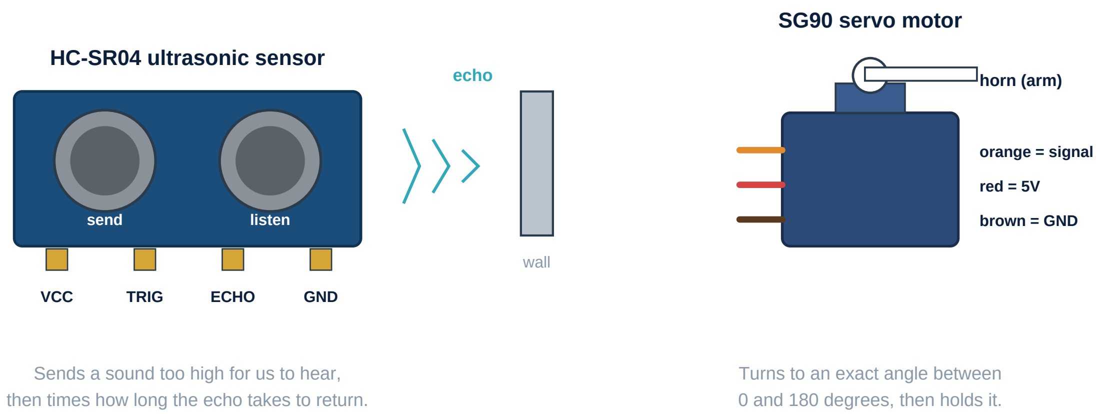


Figure 5.1 - The two new parts in your final project

Did You Know

The echo travels to the object **and back again**, so it covers double the distance. That is exactly why the formula divides by two. Bats, dolphins and submarines all use this same trick, and it has a name: **echolocation**.

12.1 What You Are Building

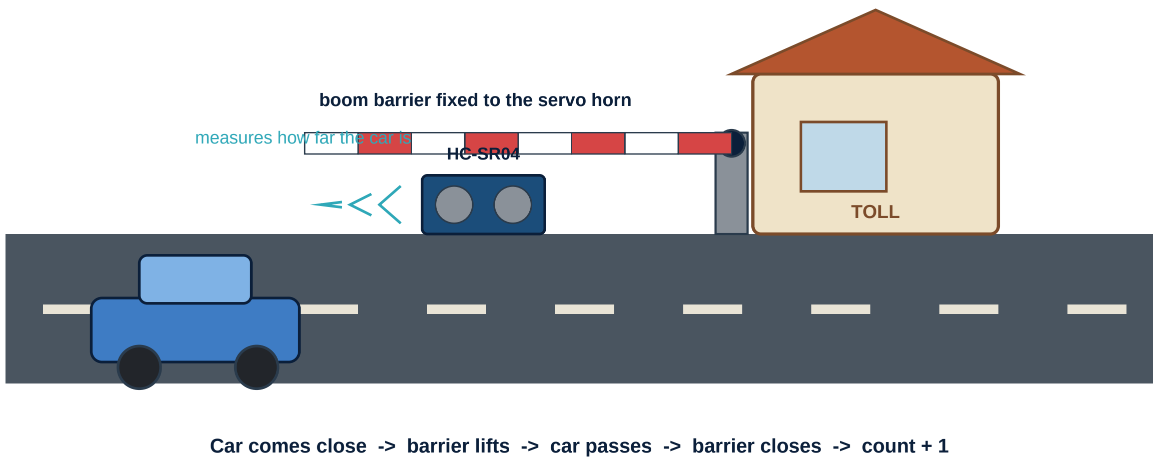


Figure 5.2 - What you are building: an automatic toll gate

Four jobs have to happen at once in this project, so decide now who is doing what. Swap roles between sessions so that everybody does everything at least once.

Role	Responsible for	Name
Builder	Cardboard booth, barrier arm, mounting the sensor firmly	________________
Wiring engineer	Every connection, and checking each one twice	________________
Programmer	Typing, uploading and changing values during testing	________________
Recorder	The notebook: test results, faults, and what fixed them	________________

SESSION 13

one class

Building and Wiring the Booth

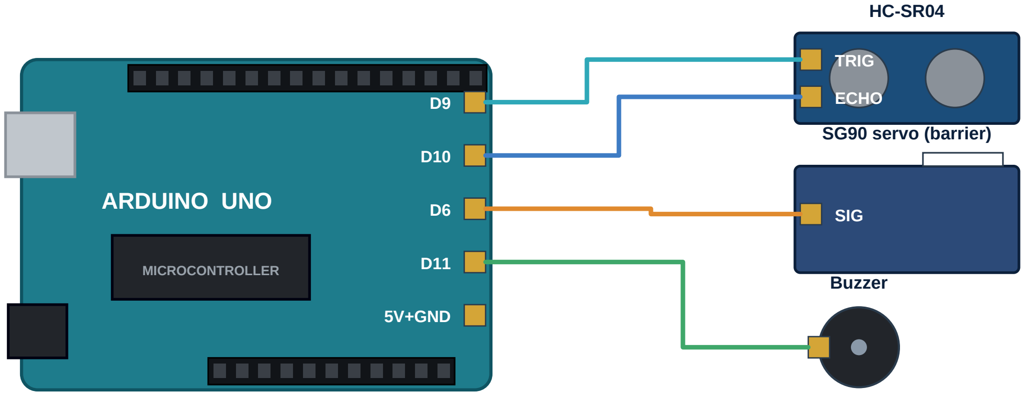
Today you will

- Build the cardboard booth and fix the barrier to the servo horn.
- Mount the ultrasonic sensor facing along the road.
- Complete every connection and check them before power on.

Build the physical model first and the wiring second. A barrier that is too heavy for the servo, or a sensor pointed at your own booth wall, will look like a code problem for an hour before you realise it was never a code problem at all.

- Keep the barrier arm light - a strip of cardboard is plenty. A heavy arm makes the servo tremble.
- Fix the servo firmly. If the servo body moves instead of the arm, nothing works properly.
- Point the HC-SR04 **along** the road, not across it, or it will see the booth instead of the car.
- Mount the sensor at roughly the height of your toy car, not above it.

Smart toll booth - connection map



Every module also needs 5V and GND. Share them through the breadboard power rails.

Figure 5.3 - One sensor in, three things out

Component	Arduino pin
HC-SR04 TRIG	D9
HC-SR04 ECHO	D10
Servo signal (orange wire)	D6
Buzzer +	D11
PASS lamp (blue LED + 220 ohm)	D12
STOP lamp (orange LED + 220 ohm)	D13
All VCC and red wires	5V
All GND, brown and black wires	GND

Watch Out

The servo, the sensor, the buzzer and both lamps all need 5V and GND. The Arduino has only a few of each, so run one wire from 5V and one from GND to the breadboard power rails, and take everything else from there.

SESSION 14

one class

The Program

Today you will

- Read and type the longest program in this book.
- Learn how to write your own instruction.
- Upload, and get the barrier moving for the first time.

This program is longer than anything you have written so far, so read it in pieces rather than all at once. There is one genuinely new idea in it: we have written **our own instruction**, called `readDistance()`. Everything needed to measure distance is tucked inside it, so the main loop stays short and readable. Professional programmers do this constantly - it is how large programs stay understandable.

SmartTollBooth.ino

```
#include <Servo.h>          // borrow ready-made servo instructions

Servo barrier;
int trig = 9;
int echo = 10;
int buzzer = 11;
int passLed = 12;
int stopLed = 13;
int count = 0;              // vehicles counted so far

void setup() {
  barrier.attach(6);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(passLed, OUTPUT);
  pinMode(stopLed, OUTPUT);
  barrier.write(0);         // start with the barrier closed
  digitalWrite(stopLed, HIGH);
  Serial.begin(9600);
}

int readDistance() {        // our own instruction
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);
  long t = pulseIn(echo, HIGH);
  return t * 0.034 / 2;     // time turned into centimetres
}

void loop() {
  int d = readDistance();

  if (d > 0 && d < 15) {    // a vehicle is close
    digitalWrite(stopLed, LOW);
    digitalWrite(passLed, HIGH);
    digitalWrite(buzzer, HIGH);
```

```
      barrier.write(90);      // lift the arm
      delay(3000);            // hold it open
      digitalWrite(buzzer, LOW);
      barrier.write(0);       // close it again
      digitalWrite(passLed, LOW);
      digitalWrite(stopLed, HIGH);

      count = count + 1;
      Serial.print("Vehicles today: ");
      Serial.println(count);
      delay(1000);            // stops one car being counted twice
    }
    delay(100);
  }
```

Two details are worth pausing on. The symbol && means *and*, so the barrier lifts only when the distance is greater than 0 **and** less than 15. And `count = count + 1;` means *take the number in count, add one, and put it back* - which is how a machine remembers something across time.

SESSION 15

one class

Testing and Your Demonstration

Today you will

- Find and fix the faults in your own machine.
- Pass every acceptance test.
- Present your project in three minutes.

Almost nothing works perfectly the first time, and that is expected. Here are the five faults that appear most often, and what actually causes each one.

Problem you may see	Most likely cause	What to do
Barrier trembles and buzzes	The servo is pulling more current than the port can give	Use a laptop USB port, not a weak phone charger
The same car is counted many times	The loop runs far faster than the car moves	Increase the delay after counting
Distance always reads 0	TRIG and ECHO are swapped	Swap D9 and D10, then test again
Barrier opens with nothing there	The sensor is aimed at the booth wall	Point it along the road, not across it
Nothing happens at all	Missing shared GND	One GND wire must reach every module

Acceptance test	Result	Tick
Barrier lifts within 1 second of a car at 15 cm	____________	■
Barrier stays open for about 3 seconds	____________	■
Barrier closes fully every single time	____________	■
The counter increases exactly once per vehicle	____________	■
Ten cars in a row with no failure	____________	■

15.1 Your Three-Minute Demonstration

Plan your three minutes. Practise it once before you stand up.

1 **The problem -** one sentence on what your machine solves.

2. **How it senses -** hold up the HC-SR04 and explain the echo in your own words.
3. **How it decides -** show the line with the number 15 in it and say what it means.
4. **How it acts -** run it live with a toy car. Let the barrier speak for you.
5. **What went wrong -** name one fault you found and fixed. Judges like this part most.
6. **What next -** one honest idea for improving it.

> **Challenge**
> Add the push switch as a 'toll paid' button so the barrier lifts only after payment is confirmed. Then print the money collected next to the vehicle count, assuming 20 rupees per vehicle.

New word	What it means
Ultrasonic	Sound too high for human ears to hear.
Echolocation	Finding distance by timing an echo.
Servo	A motor that turns to an exact angle and holds it.
Function	An instruction you write yourself, such as readDistance().

CHECK WHAT YOU LEARNT

Check Yourself - Unit 5

A. Choose the correct answer

1. The HC-SR04 measures distance by
   (a) seeing light (b) timing an echo (c) touching the object (d) weighing it
2. The distance formula divides by 2 because
   (a) it is easier (b) sound travels there and back (c) the sensor is slow (d) of the resistor
3. barrier.write(90) tells the servo to
   (a) spin forever (b) turn to 90 degrees (c) stop working (d) wait 90 seconds
4. readDistance() in this program is
   (a) a built-in word (b) an instruction we wrote (c) a mistake (d) a comment
5. The delay after counting a vehicle prevents
   (a) overheating (b) counting the same car twice (c) servo damage (d) code errors
6. The symbol && in an if statement means
   (a) or (b) and (c) not (d) equals

B. Fill in the blanks

1 A servo motor turns to an exact ______________ and holds it there.

2 The signal wire of an SG90 servo is usually ______________ in colour.

3 Sound travels roughly ______________ centimetres every millisecond.

4 count = count + 1; is how a machine ______________ something across time.

C. Answer in one or two lines

1 Explain echolocation to someone in Class 4, in two sentences.

2 Your barrier opens but never closes. Name two things you would check.

3 Name one useful thing a real toll company could do with the vehicle count.

4 Which of the three projects taught you the most, and why?